Topic 12 – Design methods, project management and product life cycle

12.1 User-centred design, circular economy and systems thinking

Spec: User-centred design: framework process, problem solving, user needs, wants and values, limitations of end user consideration; Circular economy – biologically-based systems and an understanding of how waste and pollution can be eliminated; Systems thinking – the influence of systems on commercial activity to enable all elements of a manufacturing enterprise to work together

Key facts

• **Four stages of the UCD framework?** 1 Understand context of use; 2 Specify user requirements; 3 Produce design solutions; 4 Evaluate against requirements – repeat iteratively.
• **How does UCD support problem solving?** Users help define the real problem; ideas are prototyped and tested with users; feedback guides each iteration towards a solution that meets needs.
• **Limitations of end user consideration?** Users can't always express needs or imagine new tech; samples may be unrepresentative; conflicting needs between groups; costly and slow; may stifle radical innovation; ethics/privacy of research; some users (babies) can't give feedback.
• **Three principles of the circular economy (Ellen MacArthur Foundation)?** Eliminate waste and pollution; circulate products and materials at their highest value; regenerate nature.
• **Linear vs circular economy?** Linear: take, make, dispose. Circular: design out waste, keep resources in use through repair, reuse, remanufacture and recycling, return biological materials safely to nature.
• **Business models in a circular economy?** Product-as-a-service/leasing, take-back and buy-back schemes, refurbishment and remanufacture, sharing platforms, repair services, modular upgradeable products.
• **What is systems thinking in manufacturing?** Viewing design, purchasing, production, logistics, sales, finance and customers as one connected system with feedback, so decisions benefit the whole enterprise.
• **Examples of systems that link a manufacturing enterprise?** ERP, PDM/PLM, MPS, JIT/Kanban, CIM, supply chain management, TQM, concurrent engineering.

Key terms

UCD framework process	ISO 9241-210 cycle: understand and specify the context of use → specify user requirements → produce design solutions → evaluate designs against requirements → iterate until requirements are met.
Iterative design	Repeated cycles of designing, prototyping, testing and refining.
Linear economy	Take → make → dispose: materials extracted, made into products and thrown away.
Circular economy	An economic system that designs out waste and pollution, keeps products and materials in use at their highest value, and regenerates natural systems.
Biological cycle	In the circular economy, renewable, biodegradable materials (food, wood, cotton) are safely returned to nature via composting/anaerobic digestion to regenerate soil.
Technical cycle	Finite materials (metals, polymers) are kept circulating through maintain, reuse/share, refurbish/remanufacture and recycle loops.
Cradle to cradle	Design approach (McDonough & Braungart) where all materials are either safe biological nutrients or technical nutrients that are continuously reused – no waste.
Systems thinking	Seeing an organisation as a whole system of interconnected parts (inputs, processes, outputs, feedback) so that all elements work together, rather than optimising parts in isolation.
Feedback loop	Output information is fed back to adjust inputs/processes – e.g. sales data adjusting production.

12.2 Project management strategies

Spec: Critical path analysis – the handling of complex and time sensitive operations; Scrum – how flexible, holistic product development is achieved; Six Sigma – the improvement of output quality of a process by identifying and removing the causes of defects and setting value targets of: reduce process cycle time, reduce pollution, reduce costs, increase customer satisfaction, increase profits

Key facts

• **Features of critical path analysis?** Breaks a project into activities with durations, shows dependencies and order, identifies the critical path (longest route, zero float), float on non-critical tasks, key dates; allows concurrent tasks and resource planning.
• **How do you find the critical path?** Forward pass to find earliest start times, backward pass for latest finish times; activities where EST = LFT − duration (zero float) form the critical path.
• **Three roles in a scrum team?** Product owner, scrum master, development team (developers/designers).
• **Key scrum events?** Sprint planning, daily scrum (stand-up), sprint review, sprint retrospective – within a time-boxed sprint.
• **Scrum values?** Commitment, courage, focus, openness, respect.
• **How does scrum achieve flexible, holistic development?** Cross-functional self-organising team works on the whole product together; short sprints allow constant feedback and changes in priorities; iterative increments.
• **Six Sigma value targets (spec)?** Reduce process cycle time, reduce pollution, reduce costs, increase customer satisfaction, increase profits.
• **What does 'six sigma' mean statistically?** Process variation is so small that six standard deviations fit between the mean and the nearest specification limit – about 3.4 defects per million.
• **Six Sigma belts?** Yellow, green, black and master black belt – trained staff who lead improvement projects.

Key terms

Critical path analysis (CPA)	A planning method that maps all activities, their durations and dependencies as a network to find the longest sequence of dependent activities – the critical path – which sets the minimum project time.
Critical path	The sequence of activities with zero float; any delay to them delays the whole project.
Float (slack)	The amount of time a non-critical activity can be delayed without delaying the project. Total float = latest finish − earliest start − duration.
EST / LFT	Earliest start time (forward pass) and latest finish time (backward pass) recorded at each node.
Scrum	An agile, flexible, holistic framework where small self-organising teams deliver work in short time-boxed sprints with constant feedback and adaptation.
Sprint	A time-boxed iteration (usually 1–4 weeks) producing a usable increment.
Product backlog	Prioritised list of everything that might be needed in the product, owned by the product owner.
Six Sigma	A data-driven quality method (developed by Motorola) that reduces process variation and defects to no more than 3.4 defects per million opportunities, using DMAIC.
DMAIC	Define, Measure, Analyse, Improve, Control – the Six Sigma improvement cycle.
DPMO	Defects per million opportunities – the measure of Six Sigma performance (6σ = 3.4 DPMO).

> **Exam tip (2024 Q4c):** "Reducing costs" was given in the stem – repeating it, or "reducing defects", earned no marks. Learn all five value targets.

12.3 Product life cycle

Spec: The cost, sales, profit and market implications to the designer and manufacturer of the stages of a product's life cycle: Introduction, Growth, Maturity, Decline

Key facts

• **Costs and profit at introduction?** Very high costs – R&D, tooling, start-up, training, heavy advertising; low output/high unit cost; slow sales; profit low or negative.
• **Growth stage implications?** Sales rise quickly, economies of scale lower unit cost, development costs recovered, profit rises, competitors enter, product updated from feedback, prices can be adjusted for competitive edge.
• **Maturity stage implications?** Sales peak, saturation, competitors' products, price reductions, highest profits but margins squeezed, extension strategies, plan replacements.
• **Decline stage implications?** Falling sales and profits, cut marketing, clear stock at lower prices, eventual withdrawal, still supply spare parts/support.
• **Extension strategies to keep market share in maturity?** Facelifts/model-year updates, new features/technology, include extras as standard, new markets/uses, advertising and social

media campaigns, loyalty schemes, price cuts/bundles, software upgrades.

• **Why manage a portfolio of products at different stages?** Profits from mature products fund the development of new ones, so the company always has products in growth as others decline.

Stage	Sales	Costs	Profit	Market and strategy
Introduction	Low, slow growth	Very high: R&D, design and development, tooling and machinery, training, market research, heavy advertising and promotion; high unit cost (low volume)	Low or negative (loss)	Few competitors; attract early adopters; build distribution; may use premium (skimming) or penetration pricing
Growth	Rising rapidly	Unit costs fall (economies of scale); development costs recovered	Rising	Increasing awareness and market share; competitors notice and enter; product tweaked from feedback; expand distribution; price for competitive edge
Maturity	Peak, levelling off	Low unit costs; spending on promotion to defend share	Highest total profit but margins squeezed by competition	Market saturation; rival products; price adjustments; **extension strategies** (facelifts, new features, updated technology, new markets, loyalty schemes, campaigns); prepare replacement
Decline	Falling	Reduce marketing; possible costs of withdrawal and spare-parts support	Falling, eventually loss	Outdated or replaced by new technology; clear stock with discounts; withdraw; sometimes niche/retro relaunch

Key terms

Product life cycle (PLC)	The stages a product passes through in the market – introduction, growth, maturity and decline – shown by a sales curve over time.
Introduction stage	Launch: high costs (R&D, tooling, marketing), low sales, low or negative profit, small market share, few competitors.
Growth stage	Rapid sales growth, economies of scale, rising profit, competitors enter, product refinements.
Maturity stage	Sales peak and level off, market saturation, strong competition, highest total profits but price pressure; extension strategies used.
Decline stage	Sales and profits fall as the product becomes outdated or replaced; prices cut, marketing reduced, product withdrawn.
Extension strategy	Action to prolong the maturity stage – facelifts, new features, new markets, price changes, re-branding, promotions.
Early adopters	Customers who buy a new product soon after launch – often enthusiasts willing to pay more.
Market saturation	Most potential customers already own the product, so sales growth stops.

Exam tip: Read which stage is asked for. 2022 Q4d asked about **all** stages (9 marks); 2025 Q4b asked only about the **introduction** stage (writing about all four lost marks) and 4d about maintaining share in **maturity**.